

Лабораторная №11

Настройка безопасного удалённого доступа по SSH

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Цель работы

Приобретение практических навыков по настройке удалённого доступа к серверу с помощью SSH.

Задачи работы

- Запрет доступа для пользователя root
- Ограничение списка пользователей
- Настройка дополнительных портов
- Аутентификация по ключам
- Организация SSH-туннелей

Безопасность доступа

```
[ankomyagin@client ~]$ ssh root@server.ankomyagin.net
root@server.ankomyagin.net's password:
Web console: https://server.ankomyagin.net:9090/ or https://10.0.2.15:9090/

Last login: Sat Nov 15 11:26:34 2025
root@server:~#
```

Успешное ограничение

```
[root@server.ankomyagin.net ~]# semanage port -a -t ssh_port_t -p tcp 2022
[root@server.ankomyagin.net ~]# firewall-cmd --add-port=2022/tcp
success
[root@server.ankomyagin.net ~]# firewall-cmd --add-port=2022/tcp --permanent
success
[root@server.ankomyagin.net ~]# systemctl restart sshd
[root@server.ankomyagin.net ~]# systemctl status -l sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-11-15 11:36:38 UTC; 7s ago
     Invocation: 6ddf6dd483fed4add81abedac40e03ae6
       Docs: man:sshd(8)
            man:sshd_config(5)
    Main PID: 13335 (sshd)
      Tasks: 1 (limit: 16823)
     Memory: 1M (peak: 1.3M)
        CPU: 41ms
    CGroup: /system.slice/sshd.service
            └─13335 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Nov 15 11:36:38 server.ankomyagin.net systemd[1]: Starting sshd.service - OpenSSH server daemon...
Nov 15 11:36:38 server.ankomyagin.net (sshd)[13335]: sshd.service: Referenced but unset environment variable evaluates to an empty string:
Nov 15 11:36:38 server.ankomyagin.net sshd[13335]: Server listening on 0.0.0.0 port 2022.
Nov 15 11:36:38 server.ankomyagin.net systemd[1]: Started sshd.service - OpenSSH server daemon.
Nov 15 11:36:38 server.ankomyagin.net sshd[13335]: Server listening on :: port 2022.
Nov 15 11:36:38 server.ankomyagin.net sshd[13335]: Server listening on 0.0.0.0 port 22.
Nov 15 11:36:38 server.ankomyagin.net sshd[13335]: Server listening on :: port 22.
```

Ограничение пользователей

```
[ankomyagin@client ~]$ ssh ankomyagin@server.ankomyagin.net
t
ankomyagin@server.ankomyagin.net's password:
Web console: https://server.ankomyagin.net:9090/ or https://10.0.2.15:9090/

Last login: Sat Nov 15 10:46:24 2025
[ankomyagin@server.ankomyagin.net ~]$ |
```

```
Nov 15 11:30:41 server.ankomyagin.net sshd-session[13153]: pam_unix(sshd:session): session opened for user ankomyagin(uid=1001) by ankomyagin(uid=0)
Nov 15 11:30:42 server.ankomyagin.net systemd[1]: Starting systemd-hostnamed.service - Hostname Service...
   Subject: A start job for unit systemd-hostnamed.service has begun execution
   Defined-By: systemd
   Support: https://wiki.rockylinux.org/rocky/support

   A start job for unit systemd-hostnamed.service has begun execution.

   The job identifier is 14395.
Nov 15 11:30:42 server.ankomyagin.net systemd[1]: Started systemd-hostnamed.service - Hostname Service.
```

Запрет после настройки

```
[ankomyagin@client ~]$ ssh vagrant@server.ankomyagin.net
vagrant@server.ankomyagin.net's password:
Permission denied, please try again.
vagrant@server.ankomyagin.net's password:
Last failed login: Sat Nov 15 11:38:39 UTC 2025 from 192.168.1.30 on
ssh:notty
There was 1 failed login attempt since the last successful login.
Last login: Sat Nov 15 10:56:31 2025 from 192.168.1.30
vagrant@server:~$ sudo -i
root@server:~#
logout
vagrant@server:~$
logout
Connection to server.ankomyagin.net closed.
[ankomyagin@client ~]$ ssh -p2022 vagrant@server.ankomyagin.net
vagrant@server.ankomyagin.net's password:
Last login: Sat Nov 15 11:38:45 2025 from 192.168.1.30
vagrant@server:~$ sudo -i
root@server:~#
logout
vagrant@server:~$
logout
Connection to server.ankomyagin.net closed.
[ankomyagin@client ~]$
```

Настройка портов

Ошибка привязки порта

```
[ankomyagin@client ~]$ lsof | grep TCP
lsof: WARNING: can't stat() fuse.portal file system /run/user/42/doc
Output information may be incomplete.
[ankomyagin@client ~]$ ssh -fNL 8080:localhost:80 ankomyagin@server.ankomyagin.net
[ankomyagin@client ~]$ lsof | grep TCP
lsof: WARNING: can't stat() fuse.portal file system /run/user/42/doc
Output information may be incomplete.
ssh      7906                ankomyagin      3u      IPv4    35401          0t0      TCP client.ankomyagin.net:51508->ns.ankomyagin.net:ssh (ESTABLISHED)
ssh      7906                ankomyagin      4u      IPv6    35406          0t0      TCP localhost:webcache (LISTEN)
ssh      7906                ankomyagin      5u      IPv4    35407          0t0      TCP localhost:webcache (LISTEN)
[ankomyagin@client ~]$ curl localhost:8080
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>301 Moved Permanently</title>
</head><body>
<h1>Moved Permanently</h1>
<p>The document has moved <a href="https://localhost:8080/">here</a>.</p>
</body></html>
[ankomyagin@client ~]$ |
```

Успешная работа двух портов

```
[ankomyagin@client ~]$  
ssh ankomyagin@server.ankomyagin.net hostname  
server.ankomyagin.net  
[ankomyagin@client ~]$ ssh ankomyagin@server.ankomyagin.net ls -Al  
total 20  
-rw-----. 1 ankomyagin ankomyagin 2785 Nov 15 10:56 .bash_history  
-rw-r--r--. 1 ankomyagin ankomyagin  18 Oct 29  2024 .bash_logout  
-rw-r--r--. 1 ankomyagin ankomyagin 144 Oct 29  2024 .bash_profile  
-rw-r--r--. 1 ankomyagin ankomyagin  589 Sep 18 11:08 .bashrc  
drwx-----. 5 ankomyagin ankomyagin  56 Sep 18 11:02 .cache  
drwx-----. 8 ankomyagin ankomyagin 174 Sep 18 11:02 .config  
drwxr-xr-x. 2 ankomyagin ankomyagin   6 Sep 18 11:02 Desktop  
drwxr-xr-x. 2 ankomyagin ankomyagin   6 Sep 18 11:02 Documents  
drwxr-xr-x. 2 ankomyagin ankomyagin   6 Sep 18 11:02 Downloads  
drwx-----. 4 ankomyagin ankomyagin  32 Sep 18 11:02 .local  
drwx-----. 5 ankomyagin ankomyagin 4096 Nov  8 15:05 Maildir  
drwxr-xr-x. 4 ankomyagin ankomyagin  39 Sep  8 16:42 .mozilla  
drwxr-xr-x. 2 ankomyagin ankomyagin   6 Sep 18 11:02 Music  
drwxr-xr-x. 2 ankomyagin ankomyagin   6 Sep 18 11:02 Pictures  
drwxr-xr-x. 2 ankomyagin ankomyagin   6 Sep 18 11:02 Public  
drwx-----. 2 ankomyagin ankomyagin  29 Nov 15 11:44 .ssh  
drwxr-xr-x. 2 ankomyagin ankomyagin   6 Sep 18 11:02 Templates  
drwxr-xr-x. 2 ankomyagin ankomyagin   6 Sep 18 11:02 Videos  
[ankomyagin@client ~]$ ssh ankomyagin@server.ankomyagin.net MAIL=~/.Maildir/ mail  
s-nail version v14.9.24.  Type '?' for help  
/home/ankomyagin/Maildir: 3 messages 1 unread  
  1 ankomyagin@ankomyagi  2025-11-08 08:26   14/482   "Test Maildir           "  
  2 ankomyagin           2025-11-08 09:27   18/676   "from client           "  
►U  3 Super User         2025-11-08 15:05   18/626   "LMTP test             "
```

Аутентификация по ключам

Копирование SSH-ключа

```
[root@server.ankomyagin.net ~]# cd /vagrant/provision/server
[root@server.ankomyagin.net server]# mkdir -p /vagrant/provision/server/ssh/etc/ssh
[root@server.ankomyagin.net server]# cp /etc/ssh/sshd_config /vagrant/provision/server/ssh/etc/ssh/
[root@server.ankomyagin.net server]# # Переходим в каталог provision/server
cd /vagrant/provision/server
[root@server.ankomyagin.net server]# # Переходим в каталог provision/server
cd /vagrant/provision/server
[root@server.ankomyagin.net server]# # Создаем файл скрипта
touch ssh.sh
[root@server.ankomyagin.net server]# chmod +x ssh.sh
[root@server.ankomyagin.net server]# # Открываем файл для редактирования и вставляем следующий код:
cat > ssh.sh << 'EOF'
#!/bin/bash

echo "Provisioning script $0"

echo "Copy configuration files"
cp -R /vagrant/provision/server/ssh/etc/* /etc

restorecon -vR /etc

echo "Configure firewall"
firewall-cmd --add-port=2022/tcp
firewall-cmd --add-port=2022/tcp --permanent

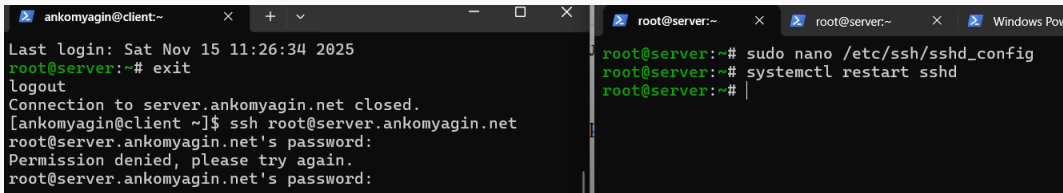
echo "Tuning SELinux"
semanage port -a -t ssh_port_t -p tcp 2022
```

```
[ankomyagin@client ~]$ ssh -YC ankomyagin@server.ankomyagin.net firefox
Warning: No xauth data; using fake authentication data for X11 forwarding.
X11 forwarding request failed on channel 0
Error: no DISPLAY environment variable specified
[ankomyagin@client ~]$
```

SSH-туннелирование

```
Nov 15 11:26:50 server.ankomyagin.net systemd[1]: Started session-10.scope - Session 10 of User root.  
Subject: A start job for unit session-10.scope has finished successfully  
Defined-By: systemd  
Support: https://wiki.rockylinux.org/rocky/support  
  
A start job for unit session-10.scope has finished successfully.  
  
The job identifier is 13641.  
Nov 15 11:26:50 server.ankomyagin.net sshd-session[13062]: pam_unix(sshd:session): session opened for user root(uid=0) by root(uid=0)  
Nov 15 11:27:20 server.ankomyagin.net systemd[1]: systemd-hostnamed.service: Deactivated successfully.  
Subject: Unit succeeded  
Defined-By: systemd  
Support: https://wiki.rockylinux.org/rocky/support  
  
The unit systemd-hostnamed.service has successfully entered the 'dead' state
```

Выполнение удалённых команд



The image shows two terminal windows side-by-side. The left window is titled 'ankomyagin@client:~' and shows the end of an SSH session from a client to a server. The right window is titled 'root@server:~' and shows commands being executed on the server.

```
ankomyagin@client:~  
Last login: Sat Nov 15 11:26:34 2025  
root@server:~# exit  
logout  
Connection to server.ankomyagin.net closed.  
[ankomyagin@client ~]$ ssh root@server.ankomyagin.net  
root@server.ankomyagin.net's password:  
Permission denied, please try again.  
root@server.ankomyagin.net's password:
```

```
root@server:~# sudo nano /etc/ssh/sshd_config  
root@server:~# systemctl restart sshd  
root@server:~#
```

Графические приложения

Ошибка X11 forwarding

```
[ankomyagin@client ~]$ ssh ankomyagin@server.ankomyagin.net
t
ankomyagin@server.ankomyagin.net's password:
Web console: https://server.ankomyagin.net:9090/ or https://10.0.2.15:9090/

Last login: Sat Nov 15 10:46:24 2025
[ankomyagin@server.ankomyagin.net ~]$ |
```

```
Nov 15 11:30:41 server.ankomyagin.net sshd-session[13153]: pam_unix(sshd:session): session opened for user ankomyagin(uid=1001) by ankomyagin(uid=0)
Nov 15 11:30:42 server.ankomyagin.net systemd[1]: Starting systemd-hostnamed.service - Hostname Service...
   Subject: A start job for unit systemd-hostnamed.service has begun execution
   Defined-By: systemd
   Support: https://wiki.rockylinux.org/rocky/support

   A start job for unit systemd-hostnamed.service has begun execution.

   The job identifier is 14395.
Nov 15 11:30:42 server.ankomyagin.net systemd[1]: Started systemd-hostnamed.service - Hostname Service.
```

Автоматизация развертывания

Копирование конфигураций

```
ankomyagin@client:~$ ssh ankomyagin@server.ankomyagin.net
ankomyagin@server.ankomyagin.net's password:
Permission denied, please try again.
ankomyagin@server.ankomyagin.net's password:
```

```
Nov 15 11:32:41 server.ankomyagin.net systemd[1]: Removed slice user-1001.slice - User Slice of UID 1001.
Subject: A stop job for unit user-1001.slice has finished
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

A stop job for unit user-1001.slice has finished.

The job identifier is 14696 and the job result is done.
Nov 15 11:32:41 server.ankomyagin.net systemd[1]: user-1001.slice: Consumed 1.223s CPU time, 11.1M memory peak.
Subject: Resources consumed by unit runtime
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

The user-1001.slice reached and consumed the indicated resources.
```

```
#!/bin/bash
echo "Provisioning script $0"
cp -R /vagrant/provision/server/ssh/etc/* /etc
firewall-cmd --add-port=2022/tcp --permanent
semanage port -a -t ssh_port_t -p tcp 2022
systemctl restart sshd
```

Контрольные вопросы

Запрет root и разрешение alice

`PermitRootLogin no`

`AllowUsers alice`

Настройка нескольких портов

Port 22

Port 2022

Port 2222

Параметры SSH-туннеля

- -f — фоновый режим
- -N — не выполнять команду
- -L — локальная переадресация

Локальная переадресация порта

```
ssh -L 5555:server2.example.com:80 user@gateway
```

Настройка SELinux для порта 2022

```
semanage port -a -t ssh_port_t -p tcp 2022
```

Разрешение порта в firewall

```
firewall-cmd --add-port=2022/tcp --permanent
```

Выводы

- Настроен безопасный доступ по SSH
- Запрещён вход для root
- Реализована аутентификация по ключам
- Настроен дополнительный порт
- Созданы скрипты автоматизации
- Повышена безопасность сервера